

SHISHIR IYER

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PROFESSIONAL SUMMARY

Backend software engineer with one year of production-quality development experience and UCSD Computer Science M.S. candidate specializing in high-performance distributed systems, cloud infrastructure, and data engineering. Proven track record delivering measurable performance improvements—including 40% latency reduction and 90% token usage reduction—across large-scale production environments. Experienced with AWS, Neo4j, Java, and generative AI workflows. Seeking a full-time Software Engineering role at a high-impact technology company. Currently on hiatus from full-time work for a master's degree

TECHNICAL SKILLS

Languages: Python (PyTorch, Pandas, NumPy, Matplotlib), Java (Spring, Mockito), JavaScript, C/C++ (CUDA, MPI, SIMD Vectorization), Bash, SQL

Cloud & Infrastructure: AWS (Lambda, Athena, S3, DynamoDB, Cloudstack), Linux, REST APIs, Kafka, Flink, Gatling

Databases: Neo4j / Cypher, DynamoDB, Vector Databases

AI / ML: Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Integration, GPT

PROFESSIONAL EXPERIENCE

CSE Teaching Assistant — Algorithms | UC San Diego *Jan 2026 – Mar 2026*

- Facilitated weekly office hours and discussion sections for CSE 101 (Design & Analysis of Algorithms), supporting 100+ students per quarter
- Graded homework assignments and proctor examinations; provided targeted written feedback to improve student understanding

Backend Software Engineer I | Intuit *Mar 2025 – Sep 2025*

- Owned large AWS cloud infrastructure for a production Neo4j graph database exceeding 350M nodes and 4.6B edges
- Led migration to an optimized infrastructure stack, achieving a 25% overall performance improvement
- Conducted systematic instance-type benchmarking using Gatling, reducing query latency by 40%
- Engineered data pipelines to prune orphan nodes (95% reduction) and supernodes (40% reduction), cutting memory usage and improving latency by an additional 10%
- Integrated S3 analytics query layer to offload live database reads and improve pipeline efficiency

Technologies: Java, Neo4j / Cypher, Bash, Linux, AWS Cloudstack, S3, Kafka, Flink, Gatling

Cybersecurity Engineering Intern | Intuit *Jun 2024 – Aug 2024*

- Extended generative AI security asset review platform; applied advanced prompt engineering to increase LLM output accuracy and relevance
- Refactored backend processing to asynchronous architecture, improving API response times by 20x
- Integrated Retrieval-Augmented Generation (RAG) into the AI recommendation workflow, reducing token consumption by 90%
- Expanded unit test coverage for the review portal backend by 50% using Spring and Mockito

Technologies: Java, Spring, GPT, RAG

CSE Tutor — Operating Systems & Computer Security | UC San Diego *Mar 2024 – Dec 2024*

- Tutored 100+ students per quarter in CSE 120 (Operating Systems) and CSE 127 (Computer Security)
- Held weekly office hours and graded exams

Software Engineering Intern | Intuit *Jun 2023 – Sep 2023*

- Designed and built a data ingestion pipeline to automate security asset reviews using generative AI
- Architected end-to-end flow: AWS Athena queries historical JIRA tickets → AWS Lambda → DynamoDB → Vector Database → LLM ingestion
- System projected to increase coverage of low-to-medium risk assets by 60%

Technologies: Java, Spring, DynamoDB, AWS Lambda, AWS Athena, Vector DB

EDUCATION

M.S. Computer Science — UC San Diego *Sep 2025 – Jun 2026 (Expected)*

Relevant Coursework: Operating Systems, Parallel Computing, Computer Security, Computer Architecture, Cryptanalysis, Computational Biology, Functional Genomics, Modern Cryptography, Randomized Algorithms, Machine Learning for Music

B.S. Computer Science — UC San Diego *Sep 2021 – Mar 2025*

Relevant Coursework: Advanced Data Structures, Algorithm Design & Analysis, Database Systems, Deep Learning, Computer Vision, Compiler Construction, Parallel Computing, Cryptography, Computer Architecture, Operating Systems, Computer Security

PROJECTS & RESEARCH

[Auto-forecast](#) | *Apr 2026 – Present*

- Built an end-to-end pipeline to generate weather forecasts from NWS forecast discussions and GFS model data
- Extracted synoptic-scale meteorological features from GRIB2 archives including surface pressure minima, 500mb height anomalies, and jet stream characteristics using XArray and MetPy
- Simplified NWS forecast discussions using a locally-run LLM (Llama 3.1) with structured output validation and hallucination detection
- Fine-tuned a 1.1B parameter small language model (TinyLlama) using LoRA on paired meteorological feature / summary examples

Technologies: *Python (PyTorch, XArray, MetPy), HuggingFace*

[MBX High-Performance Computing](#) | **Paesani Research Group, UCSD** *Feb 2024 – Present*

- Optimized the 3-body water polynomial in the MBX molecular simulation suite using SIMD vectorization across 20,000+ lines of C++ code
- Achieved a 6.9x runtime improvement, 80% memory usage reduction via liveness analysis and redundant computation elimination
- Co-authored a peer-reviewed software release paper detailing improvements: doi.org/10.1021/acs.jctc.4c01333
- Currently redesigning the particle mesh Ewald sum computation to leverage vectorization and parallelism

Technologies: *C++, Python, SIMD, MPI*

[Reconstructing \$k\$ -mer Spectra with Deep Learning](#) | **Computational Biology, UC San Diego** *Mar 2026*

- Developed a multi-layer perceptron to reconstruct the k -mer spectrum from a genome skim for rapid and inexpensive genome characterization, achieving a mean Jensen-Shannon distance (JSD) of 0.0063 for predicted repeat spectra
- Co-authored a paper detailing neural network design and results:
https://github.com/shishir03/cse280a-genome-skim/blob/main/CSE_280A_Paper.pdf

Technologies: *Python / PyTorch*

Agentic AI Payment System Security Research | **Graduate Computer Security, UC San Diego** *Dec 2025*

- Conducted security research into payment protocols enabling AI agent-driven purchases, identifying a class of prompt injection vulnerabilities
- Demonstrated that insufficient input validation in agentic payment flows can allow adversarial actors to manipulate agent behavior at the protocol level
- Wrote a blog post detailing results:
<https://medium.com/@shishir-iyer/security-considerations-for-agentic-payment-systems-02107ee00174>

Technologies: *Python, Agentic AI*

[Flap.js team Lead & Maintainer](#) | **UC San Diego** *Jan 2023 – Mar 2024*

- Led development of Flap.js, an open-source browser-based tool for designing finite automata (JFLAP clone), actively used by UCSD students
- Delivered Moore and Mealy machine support and implemented an automatic graph layout algorithm

Technologies: *JavaScript*

HONORS & AWARDS

- Intuit Scholarship Recipient (2021 – 2025) — Competitive merit-based scholarship awarded to high-achieving Computer Science students